Software Requirement Specifications

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Purpose

The purpose of this Software Requirements Specification (SRS) document is to clearly define the functional and non-functional requirements for the proposed system enhancements at Naya-Tel, based on the requirement analysis conducted through interviews with their senior members. This document will serve as a foundational guide for the development team and stakeholders to understand the scope and expected functionalities of the system.

Scope of Document

The main scope of this SRS documentation is to highlight the existing software's they are using and which are currently under development that need more light into the technologies being used such as their AI report generation software, Customer service portal for complaints and then also a better and reliable database with low-cost measures and estimates to enhance their revenue and annual income overall.

Problem Statement

When we conducted our interview we came through the following problems that they were currently facing for the Naya-Tel system such as:

- AI based automations
- Security related issues for their existing system
- System Design
- Points Based Rewards System – PBRS
- Performance Issues with overall system
- Scalability
- Lack of decision-making software's for development processes
- Automated Bot Training Systems based on LLM and NLM's
- Poor database schemas and execution with current system
- Low availability of servers

These were some of the problems that they are facing with their existing system and need to improve in the next coming years. The systems that they are using right now are under developed and need better integration through proper models and systems.

Existing System Analysis

When we did a proper analysis, we came to know the following things about their **As-is** and **To-be** models that we discussed:

AS-IS SYSTEM

Executive Summary

Naya-Tel's basic purpose is to become a role model of trust, set highest standards of quality and add value to the society. In other words, they also aim to provide cloud services and a soothing FTTH (Fiber-To-The-Home) for all its customers. They also aim to provide leading network services for their clients and customers aiming to reach the international market trends and then increase its scope for all the network consumers. Their signature product is the internet-based services but their focus is on cloud-based services nowadays.

a) Stakeholders Interviewed

Following are the stakeholders that we have interviewed for this project:

- HR Manager
- Deputy Manager (Software Department)
- Project Manager (Software Department)
- Product Manager
- Senior Database Administrator
- Head of Customer Service Department
- Head of Networking Department

b) Pain Points for Naya-Tel

The current pain points that Naya-Tel experiences in their existing system are the following as shown below:

- AI based automations
- Security related issues for their existing system
- System Design
- Points Based Rewards System – PBRS
- Performance Issues with overall system
- Scalability
- Lack of decision-making software's for development processes
- Automated Bot Training Systems based on LLM and NLM's

Detailed Description of Current System

This part of the document provides us with the detailed description of how their system is running now and what actions do they perform. All these actions have been determined after conducting a thorough interview with the concerned Naya-Tel members.

DD-01

Process Name: Database Integration for Naya-Tel systems
Workers: Database Administrator
Triggers: Database administrators perform CRUD operations.
Tool Used: Oracle(past) – Postgres (Current)
Problems Faced: Costly and low speed execution time.

DD-02

Process Name: Reports Generation
Workers: Technical Team
Triggers: Technical Team generates reports by hand through MS Word.
Problems Faced: Time consuming and more workload.

DD-03

Process Name: Customer Service Application Interface
Workers: Customer Service Team
Triggers: Concerned personnel from customer service team guide customers
Problems Faced: Lack of AI based automation techniques and interfaces

TO-BE SYSTEM

This part of the document clearly shows us what their new and enhanced system is going to integrate and what are going to be it's features as mentioned below:

TB-01

New and improved system: Naya-Tel Database 2.0
Purpose: Naya-Tel's natural database with proper schemas and functionalities
Performed by: Database Administrator
Advantages:

- Faster Execution statements

- Human understandable syntaxes
- Low-cost coverages
- Faster run time and integration with servers
- Improved and enhanced CRUD operations

TB-02

New and improved system: AI Client 2.0
Purpose: Generate AI based reports offering vast options
Performed by/Used by: Overall Naya-Tel staff
Advantages:

- Faster Execution Times
- Lower workload on staff
- Accuracy based revenue reports
- Accessibility enhanced

TB-03

New and improved system: Naya-Tel Customer Agent 24/7
Purpose: Provide 24/7 cloud-based interaction between customers seeking complaints
Performed by: Customer Service Department
Advantages:

- Provide vast options for customer service complaints
- Enhanced Multi-lingual support for customers
- Faster response time for customers
- Enhanced video tutorials for related issues

Elicitation Techniques Used

Elicitation techniques are a crucial part in gathering requirements in Software Engineering. With our semester project with Naya-Tel we used various elicitation techniques to analyze the requirements and then give them a shape of functional and non-functional requirements which shall be explained after we have described our elicitation techniques.

1.1 Primary Elicitation Technique – Interviews

Our main and primary elicitation technique was conducting face-to-face interviews with their team which included their HR/Executive, Project Manager, Deputy Manager (Software Team), Product Manager and then also their senior database administrator.

Justification for using Interviews as Primary

If we talk about why we thought it was best to choose the interviews as the primary elicitation technique because it provided us with the following things which enhanced our gathering requirements process.

Direct Information Source
We were able to gather the information directly from the concerned people at Naya-Tel through which we properly understood their statements and requirements.

Detailed Insights
We as a team were able to analyze their operational techniques and working of the Naya-Tel system which led us to a clearer path towards gathering the requirements for our semester project.

Follow up Questions
While we were conducting the interview, we were able to ask them follow up questions and then also cleared our misunderstandings in order to avoid unfiltered data or without any clear context.

Relationship Building
During the interview being conducted we were able to build a professional work - relation with them and forwarded our hand for future visits as well regarding any more requirements or guidance which wouldn't be possible without this professional interview.

1.2 Secondary Elicitation Technique – Process Analysis / Observation

Alongside with conducting interviews we also analyzed their processes and asked them about the way their reports work when generating them and then also asked them about the Database they have been using and how it helped them making a seamless software for their customers and nation-wide user's range. Process analysis and observation helped us

understand their internal system working and how the workflow processes were dealt with in the office-space amongst colleagues and staff members from Naya-Tel.

Justification for using Process Analysis / Observations

After proper process analysis and observations, we were able to identify the problem statements and then also the arguments for our "As-is" statement which helped us get a deeper insight into our SRS documentation.

Overall, the elicitation technique that we used clearly shows us that we were able to gather the requirements in a professional way and timely manner whilst keeping the professional relationship in mind. This is the sole purpose of the justifications for using these elicitation techniques.

Specific Requirements

After going through the elicitation techniques and interviews held, we came up with the following functional and non-functional requirements for the semester project.

Functional Requirements

These are going to be the requirements that the end user are going to be having in the newly integrated and improved enhancements for the project.

1.1 Naya-Tel Database 2.0 Enhancements

1.1.1 Data storage and retrieval

This shall allow the Database Administrator (DBA's) to store and retrieve data with less execution times as compared to the old system which they are currently using.

1.1.2 Schema Management

This system shall enable and have the basic CRUD features such as Create, Update, Read and Delete implied on all defined data entries by the DBA.

1.1.3 Integration

This will allow the system to seamlessly integrate with their existing and new application servers that will enable efficient data exchange or transfers.

1.2 AI Client 2.0

1.2.1 AI Based Report Generation

This AI based report generator will enable the staff to generate various types of reports such as revenue, operational and then performance by using our

predefined AI algorithms and some pre-existing one's as well. It shall produce accurate results with no room for errors and inaccurate data.

1.2.2 Report Accessibility and Management

The system or concerned Naya-Tel employees shall have the ability to download their reports in the form of a PDF file or even in excel in order to store the data in a tabular form for their reports and then have a logging feature as well which stores their reports for later audits or retrievals.

1.2.3 Data sources for AI

This feature shall allow the system to access Naya-Tel's database and then automatically work with the algorithm that was embedded by us for their system to work on and then some internal sources as well such as there billing systems and customer usage logs.

1.3 Naya-Tel Customer Service Portal 24/7

1.3.1 Cloud-based interactions

The system shall provide a 24/7 accessible cloud-based platform for their customers interactions which shall make the cloud operations smoother and more efficient.

1.3.2 AI Based Customer Support

The system shall enable itself to integrate information from the LLM's and NLM's that we shall train for the overall customer services and then their queries. This shall be trained on a predefined dataset from Naya-Tel regarding its active customers.

1.3.3 Video Tutorials and Self Service

This feature shall allow the user to have a 24/7 video tutorial for their customer service-related queries or any issue they are facing at that time.

1.3.4 Agent Handover

This shall allow the customers to directly communicate with one of the human representatives from the Naya-Tel customer service team and then guide them forward based on the past conversation with their AI Agent.

Non-Functional Requirements

These are going to be the quality attributes of the system that we are going to be developing for Naya-Tel and then also the functions of it.

1.1 Performance

1.1.1 Database Execution Speed

This shall enhance their current system's execution speed which shall complete the CRUD operations within 200ms for 90% of their transactions during the peak load times and operations.

1.1.2 Report Generation Time

The AI Based reports shall be generated and they shall be ready to access within 5 secs for normal reports and then 10 secs for a bit complex reports.

1.1.3 Customer Agent Response Time

Automated response from the Naya-Tel customers shall be transferred to the human agent within 10 secs of the starting time if they do not wish to proceed ahead with the AI agent. Agent shall respond to them within 10 secs of active time and then proceed with their query.

1.2 Reliability

1.2.1 System Uptime

Through their enhanced system features and modules, the Naya-Tel company shall be able to expect a total of 99.99% uptime throughout the year with an exception of 0.1% during maintenance and check period.

1.2.2 Data consistency

All the data across the database and their servers shall stay in sync with the live data that is being fed by the staff and technical responsible personnel.

1.2.3 Failover capability

This shall enable the server to deploy on another instance if any one instance of the server fails which reduces the system's downtime to the minimal and users won't have to suffer anymore.

1.3 Scalability

1.3.1 Concurrent Users

The system shall be able to support more users without having any disruptions in the server's load capacity or performance durability.

1.3.2 Data Volume

The database system will be to support handling 80% increase in its data capacity year over year for the next 5 years without having to switch to a different database architecture.

1.4 Security

1.4.1 Data Protection

All the data of the customer and staff from user will be encrypted with proper layers in the database and in the servers in order to maintain users and staff confidentiality and privacy.

1.4.2 Access Control

The system shall assign and implement a Robust Role-Based Access Control in which it shall allot the functionalities to only the concerned people and stakeholders or actors and workers.

1.4.3 Threat Protection

This system shall be resilient against common web vulnerabilities such as a SQL injection or trojan horse attack or malware attacks and enable DDoS protection for its servers and databases and applications.

1.5 Usability

1.5.1 User Interface (UI)

The interfaces for the AI Client 2.0 and Naya-Tel's Customer Agent 24/7 shall be intuitive and easy to locate for their intended users nation-wide.

1.5.2 Training

New staff members shall be able to effectively use the interface in order practice sessions about interacting with the customers

1.5.3 Evaluation Based Pointing Scheme

The system shall be able to have a evaluation based pointing scheme and give its staff a pointing scheme which shall increase its chance to employment and further promotions.

1.6 Cost-Effectiveness

1.6.1 Operational Cost Reduction

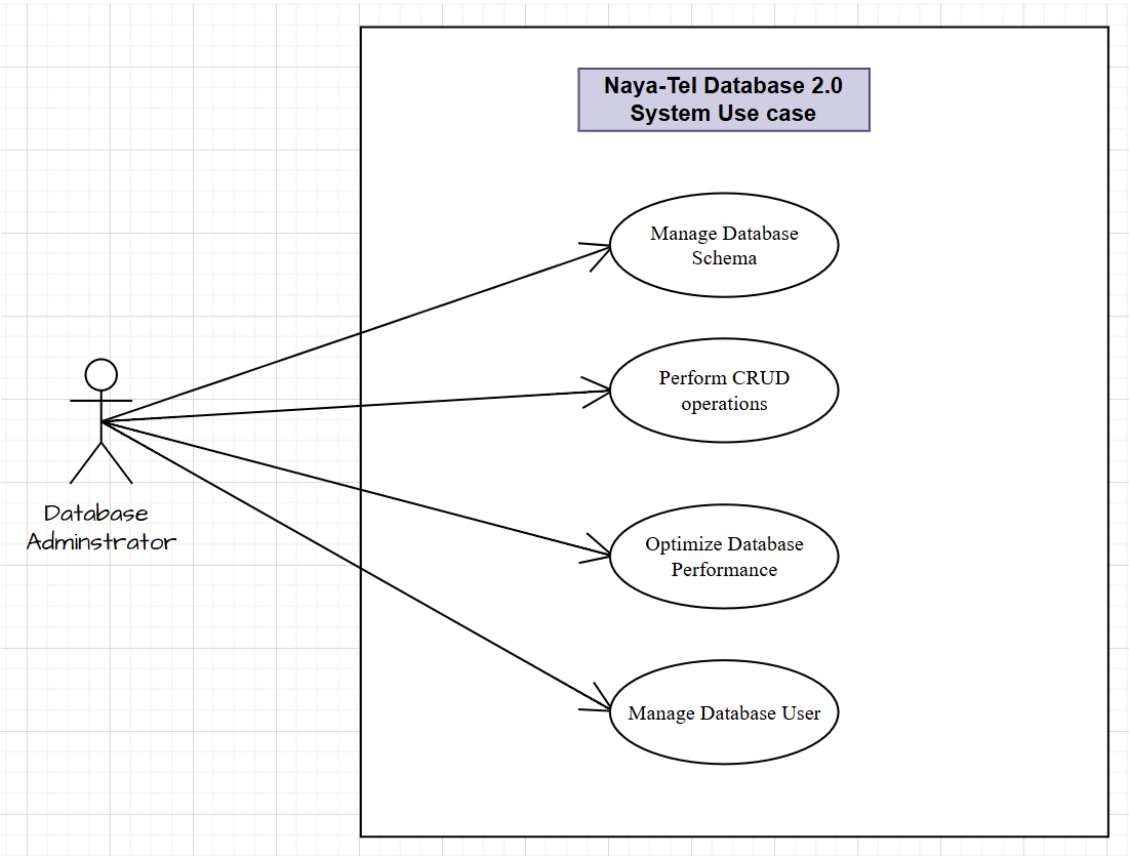
The new database system shall result in total reduction of the operational cost that they have been paying so far and bring it to a minimal. In the first year they shall come down to 15-20% of their total operational cost estimates and have a fairer balance check for their financing.

Use Case Diagrams

Based on our ***To-Is*** list we are going to mention the following use-case diagrams:

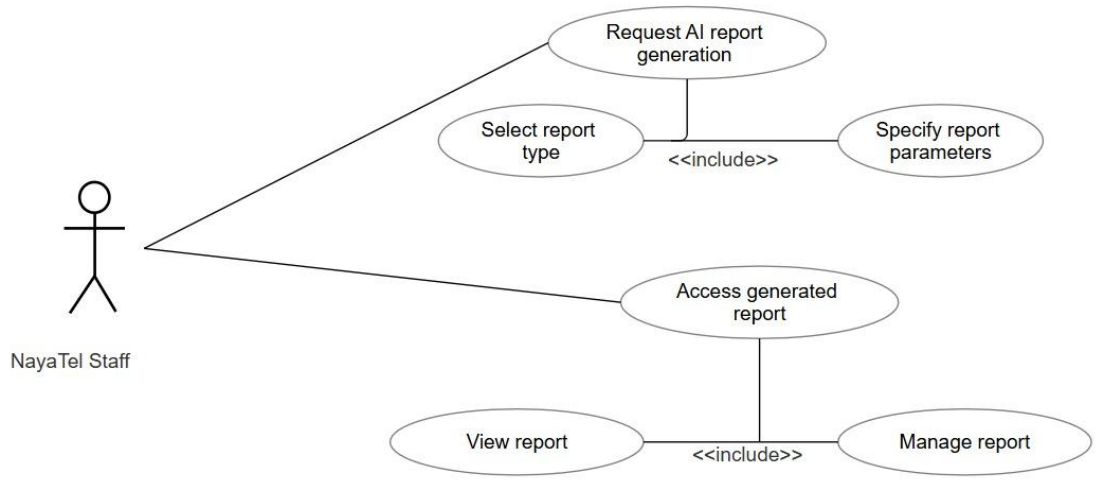
- Naya-Tel Database 2.0
- AI Client 2.0
- Naya-Tel Customer Service 24/7 Agent

Naya-Tel Database 2.0



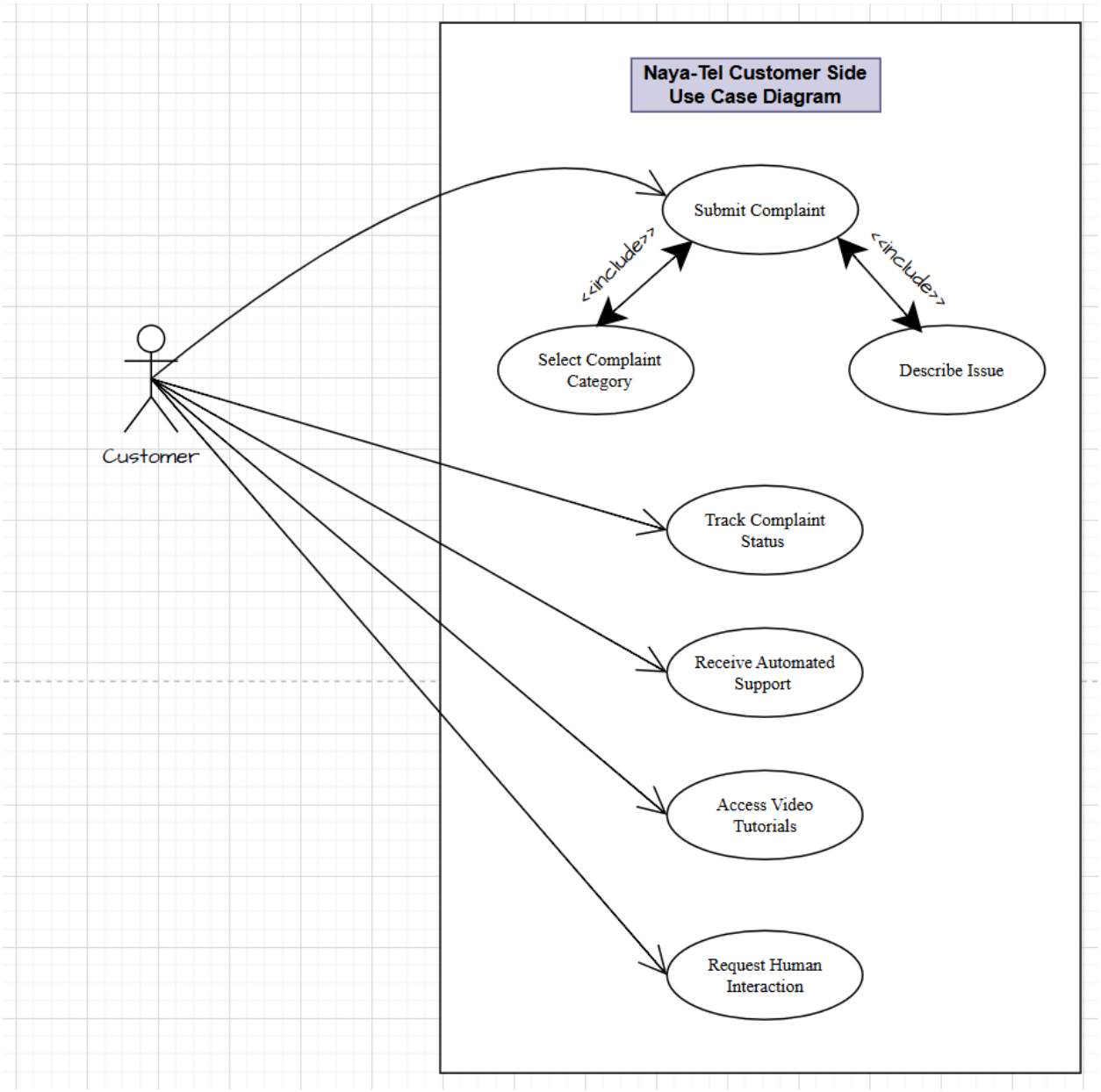
Title: Naya-Tel Database 2.0 System Use Case
Actor: Database Administrator
Number of Use-Cases: UC-001, UC-002, UC-003, UC-004
Derived Origin: TB-01

AI Client 2.0



Title: AI Client 2.0
Actor: Over-all Naya-Tell Staff
Use Cases: UC-001, UC-002, UC-003, UC-004, UC-005, UC-006
Derived Origin: TB-02

Naya-Tel Customer Agent 24/7

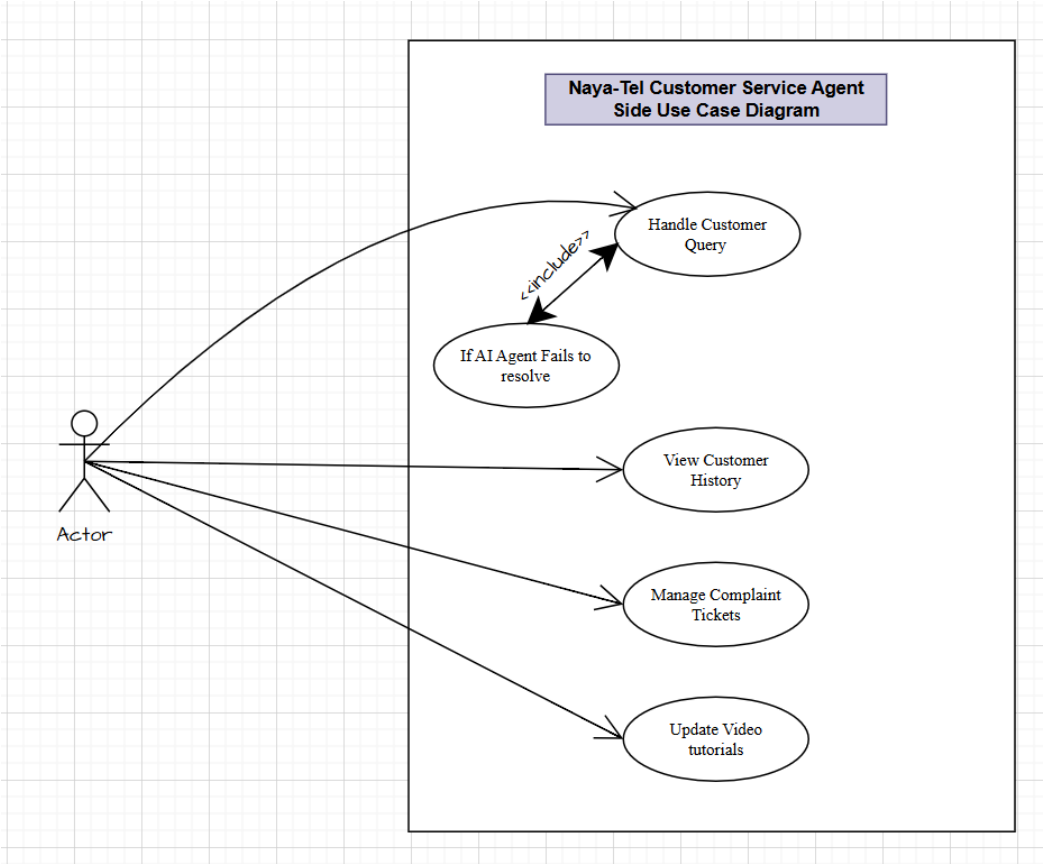


Title: Naya-Tel Customer Side

Actor: Customer

Use cases: UC-001, UC-002, UC-003, UC-004, UC-006, UC-006

Derived Origin: TB-03

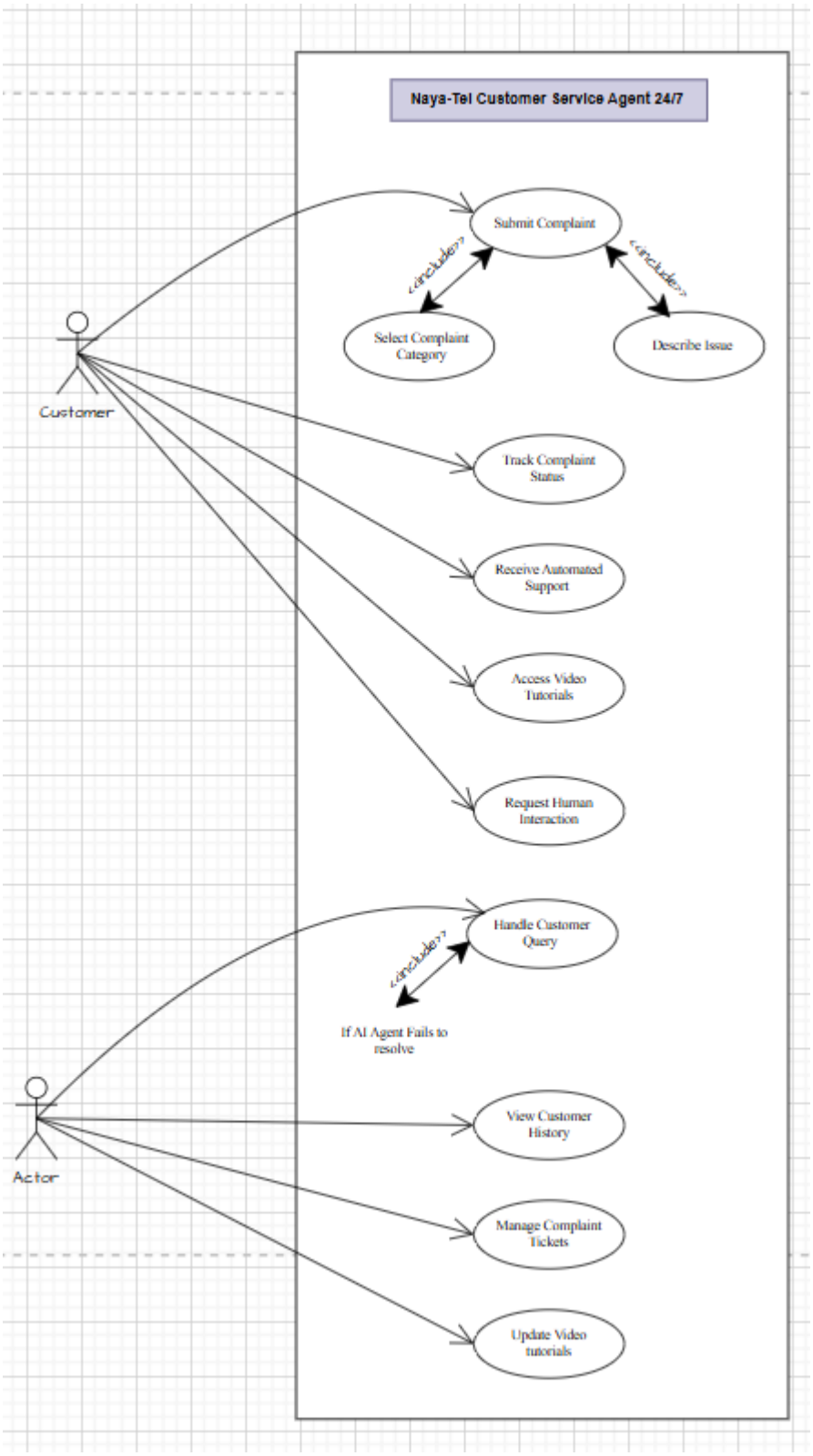


Title: Customer Service Agent Side

Actor: Customer Service Agent

Use Cases: UC-001, UC-002, UC-003, UC-004, UC-005

Derived Origin: TB-03


Naya-Tel Customer Service Agent 24/7
Submit Complaint
<<include>>
<<include>>
Select Complaint Category
Describe Issue
Customer
Track Complaint Status
Receive Automated Support
Access Video Tutorials
Request Human Interaction
Handle Customer Query
<<include>>
If AI Agent Fails to resolve
Actor
View Customer History
Manage Complaint Tickets
Update Video tutorials